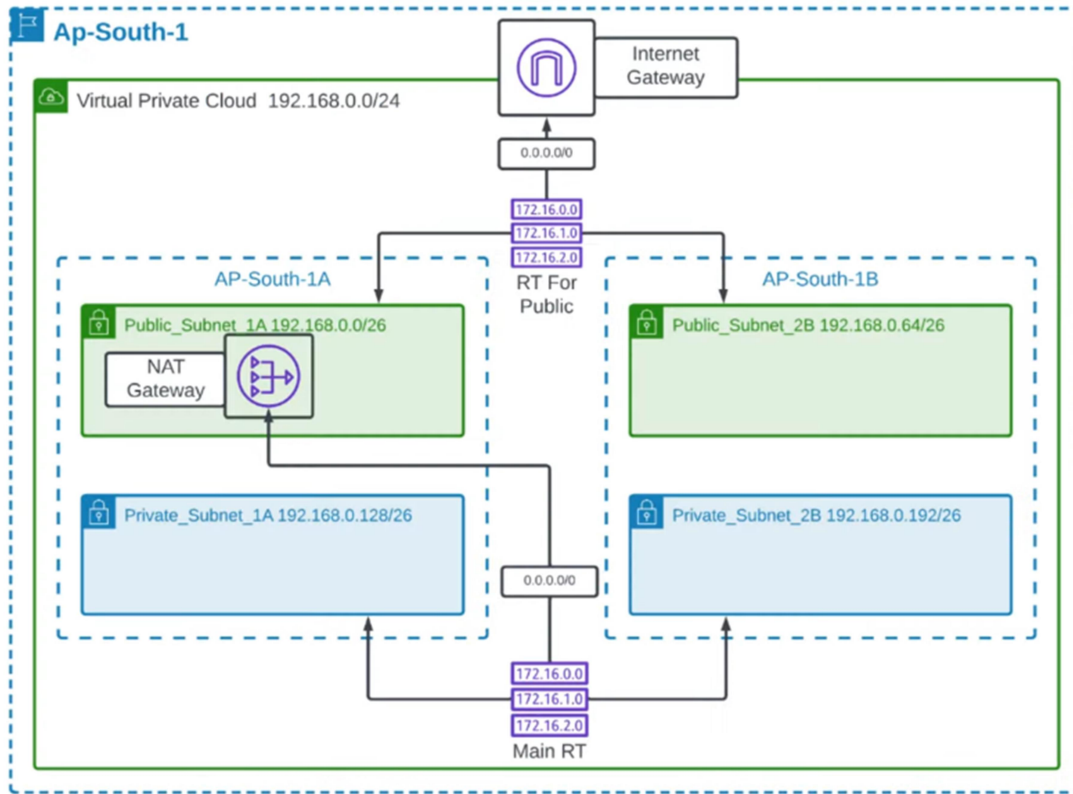




Design vpc as above diagram.

Create secu.group for alb.

Security group name [Info](#)

Alb-SG

Name cannot be edited after creation.

Description [Info](#)

Securing ALB From Internet

VPC [Info](#)

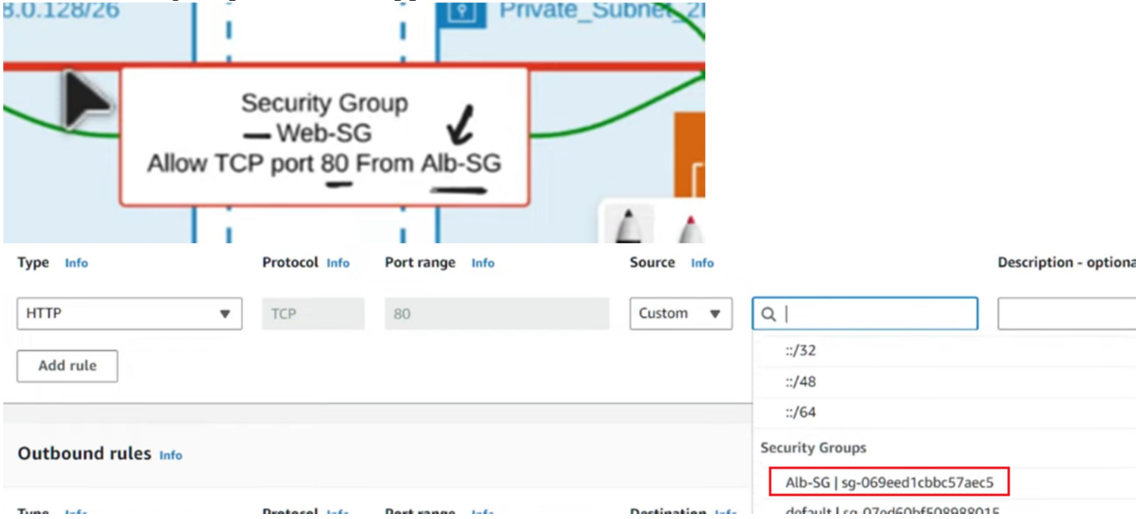
vpc-0e25a1bdc72bbd66 (MyVPC)

Inbound rules [Info](#)

Type Info	Protocol Info	Port range Info	Source Info	Description - option
HTTP	TCP	80	Anywh... Info	
			0.0.0.0/0	

[Add rule](#)

Create sec.group for web app instance



Security Group — Web-SG
Allow TCP port 80 From Alb-SG

Type: HTTP, Protocol: TCP, Port range: 80, Source: Custom

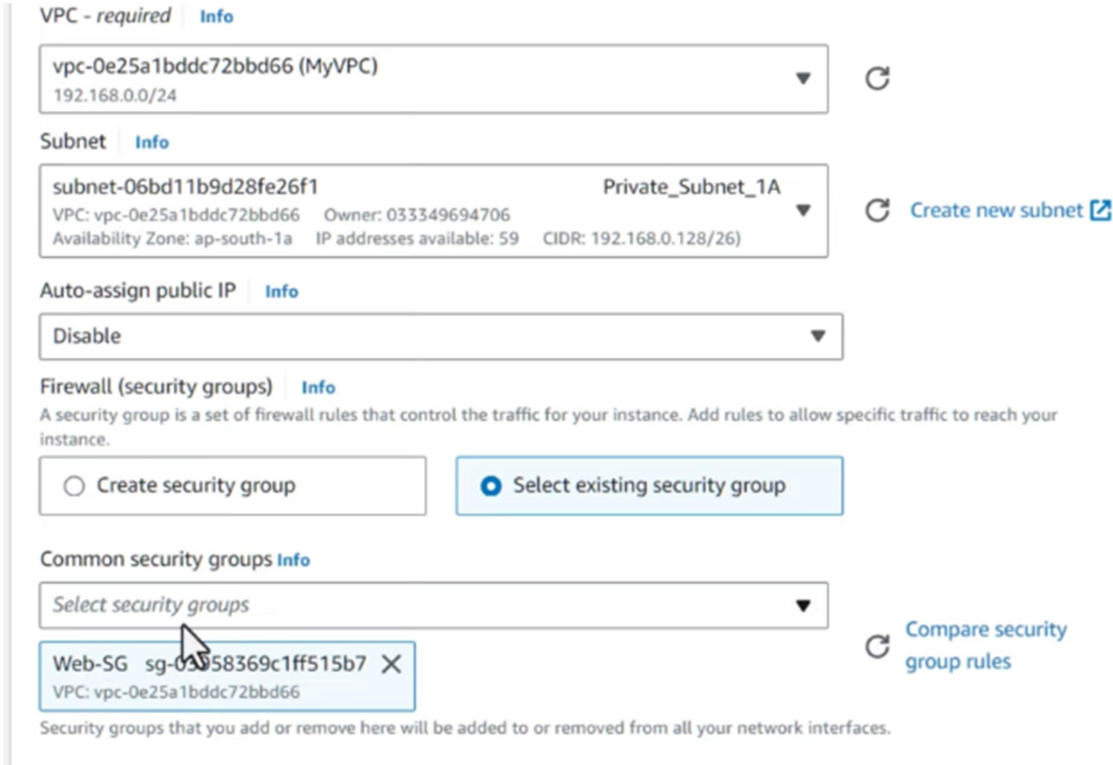
Add rule

Outbound rules

Security Groups

- Alb-SG | sg-069eed1cbbc57aec5

create instance



VPC - required: vpc-0e25a1bddc72bbd66 (MyVPC) 192.168.0.0/24

Subnet: subnet-06bd11b9d28fe26f1 Private_Subnet_1A
VPC: vpc-0e25a1bddc72bbd66 Owner: 033349694706
Availability Zone: ap-south-1a IP addresses available: 59 CIDR: 192.168.0.128/26

Auto-assign public IP: Disable

Firewall (security groups):
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group ☒ Select existing security group

Common security groups: Select security groups

Web-SG sg-069eed1cbbc57aec5
VPC: vpc-0e25a1bddc72bbd66

Compare security group rules

Security groups that you add or remove here will be added to or removed from all your network interfaces.

And in advanced detail go to user data script write below code

```
#!/bin/bash
# Install Apache Web Server and PHP
yum update -y
yum install -y httpd php

# Manually set the server name
SERVER_NAME="Web-Server1 " # Replace YOUR_SERVER_NAME with the actual
server name

# Create a PHP file with added styling
cat > /var/www/html/index.php <<EOF
<html>
```

```

<head>
  <title>Load Balancer Test</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #f0f0f5;
      color: #333;
      text-align: center;
      padding-top: 50px;
    }
    h1 {
      color: #0066cc;
      background-color: #ffff99;
      padding: 10px;
      border-radius: 8px;
      display: inline-block;
      border: 1px solid #cccccc;
      box-shadow: 3px 3px 5px #888888;
    }
    .highlight {
      color: #cc0000;
      background-color: #ffff99;
      padding: 5px;
      border-radius: 5px;
    }
    a {
      color: #cc0000;
    }
  </style>
</head>
<body>
  <h1>Welcome to CloudFolks Load Balancer Test Website</h1>
  <p><span class="highlight">You are getting this webpage from
<strong>${SERVER_NAME}</strong></span></p>
  <p>Powered by CloudFolks Hub</p>
  <p>Visit our website: <a
href="https://learn.bhaveshatara.live">https://learn.bhaveshatara.live<
/a></p>
</body>
</html>
EOF

```

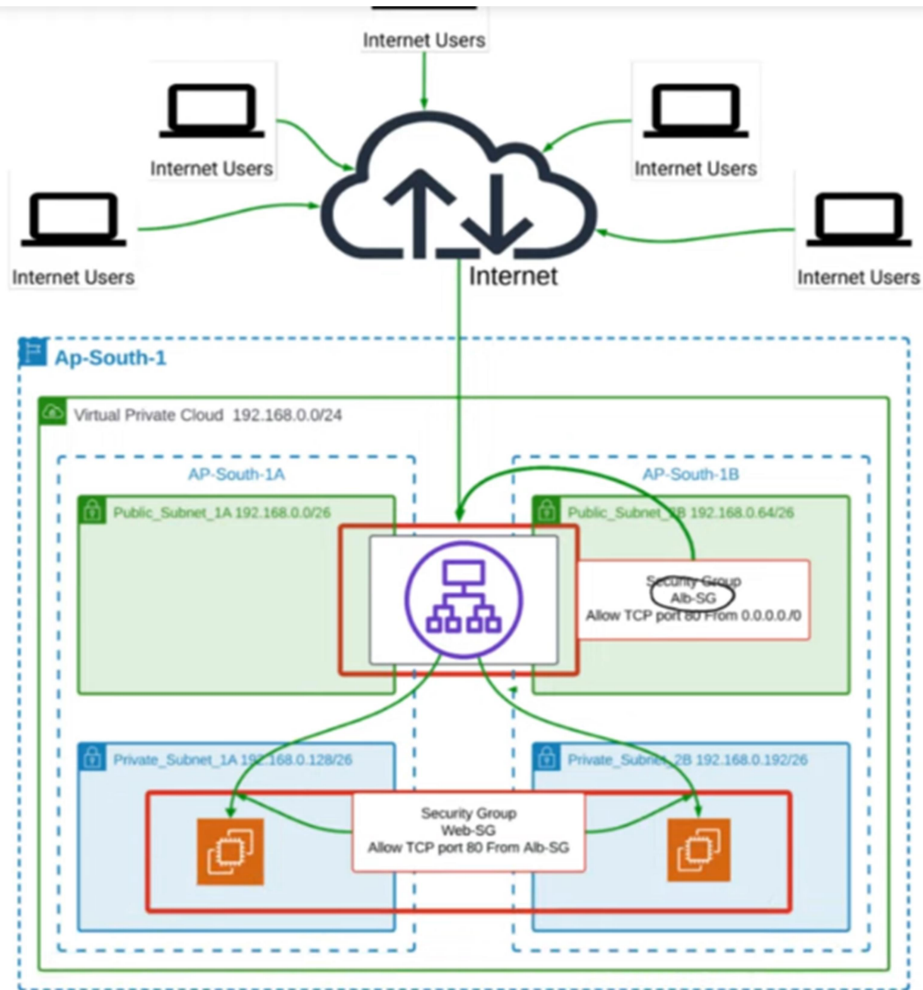
```

# Enable and start the Apache Web Server
systemctl enable httpd
systemctl start httpd

```

Same instance create one more and name web-server1 > subnet private-subnet-1B, sg-websg,userdatagram paste above code and launch instance.

Set up ALB



Go to ec2 dashboard and go to load balancer click to > Create Load Balancer > click To Application LoadBalancer

Basic configuration

Load balancer name

Name must be unique within your AWS account and can't be changed after the load balancer is created.

ALB

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme [Info](#)

Scheme can't be changed after the load balancer is created.

☒ Internet-facing

An internet-facing load balancer routes requests from clients over the internet to targets. Requires a public subnet. [Learn more](#)

☐ Internal

An internal load balancer routes requests from clients to targets using private IP addresses.

IP address type [Info](#)

Select the type of IP addresses that your subnets use.

☒ IPv4

Recommended for internal load balancers.

☐ Dualstack

Includes IPv4 and IPv6 addresses.

VPC [Info](#)

Select the virtual private cloud (VPC) for your targets or you can [create a new VPC](#). Only VPCs with an internet gateway are enabled for selection. The selected VPC can't be of the load balancer is created. To confirm the VPC for your targets, view your [target groups](#).

MyVPC
vpc-0e25a1bddc72bbd66
IPv4: 192.168.0.0/24

Mappings [Info](#)

Select at least two Availability Zones and one subnet per zone. The load balancer routes traffic to targets in these Availability Zones only. Availability Zones that are not supported by the load balancer or the VPC are not available for selection.

☒ **ap-south-1a (aps1-az1)**

Subnet
subnet-0b9acda39bc95d88b **Public_Subnet_1A**

IPv4 address
Assigned by AWS

☒ **ap-south-1b (aps1-az3)**

Subnet
subnet-04a17b8f6a72da5e0 **Public_Subnet_2B**

IPv4 address
Assigned by AWS

Security groups [Info](#)

A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can [create a new security group](#).

Security groups

Select up to 5 security groups

Alb-SG
sg-069eed1cbbc57aec5 VPC: vpc-0e25a1bddc72bbd66

Now target instances created already

Listeners and routing [Info](#)

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

Listener HTTP:80 Remove

Protocol **HTTP** **Port** 80 **Default action** [Info](#)
1-65535 Forward to Select a target group Create target group

Listener tags - optional
Consider adding tags to your listener. Tags enable you to categorize your AWS resources so you can more easily manage them.

Add listener tag
You can add up to 50 more tags.

Add listener

Choose a target type

☒ **Instances**
• Supports load balancing to instances within a specific VPC.
Facilitates the use of [Amazon EC2 Auto Scaling](#) to manage and scale your EC2 capacity.

☐ **IP addresses**

Target-1

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Protocol : Port

Choose a protocol for your target group that corresponds to the Load Balancer type that will route traffic to it. Some protocols now include anomaly detection for the targets and you can set mitigation options once your target group is created. This choice cannot be changed after creation

HTTP

80

1-65535

IP address type

Only targets with the indicated IP address type can be registered to this target group.

☒ IPv4

Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.

☐ IPv6

Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). [Learn more](#)

VPC

Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

MyVPC

vpc-0e25a1bddc72bbd66

IPv4: 192.168.0.0/24

then click next

Register targets

This is an optional step to create a target group. However, to ensure that your load balancer routes traffic to this target group you must

Available instances (2/2)

Filter instances

☒	Instance ID	Name	State	Security groups
☒	i-0f1b194e788ffbd77	Web-Server-2	Running	Web-SG
☒	i-030aca00ad04c6801	Web-Server-1	Running	Web-SG

2 selected

Ports for the selected instances
Ports for routing traffic to the selected instances.

80

1-65535 (separate multiple ports with commas)

Include as pending below

Review targets

Targets (2)

Filter targets

Show only pending

Remove	Health status	Instance ID	Name	Port	State	Security groups	Zone	Private IPv4
X	Pending	i-0f1b194e788ffbd77	Web-Server-2	80	Running	Web-SG	ap-south-1b	192.168.0.2
X	Pending	i-030aca00ad04c6801	Web-Server-1	80	Running	Web-SG	ap-south-1a	192.168.0.1

2 pending

Cancel

Previews

Again go to Load Balancer and then refresh

Protocol: HTTP

Port: 80

1-65535

Default action: Forward to Target-1

Target type: Instance, IPv4

HTTP

1

Create target group

then click to create load balancer

☐	Name	DNS name	State
☐	ALB	ALB-1630058658.ap-south-1.elb.amazonaws.com	Active

copy dns and paste to browser

Finish and check on browser server status by load balancer

If want to use domain name instead of load balancer dns then go to route53 and do setup

Route 53 > Hosted zones > cloudfolk.in > Create record

Create record Info

Quick create record [Switch to wizard](#)

▼ Record 1 Delete

Record name Info
test .cloudfolk.in
Keep blank to create a record for the root domain.

Record type Info
A – Routes traffic to an IPv4 address and some AWS resources ▼

☒ Alias

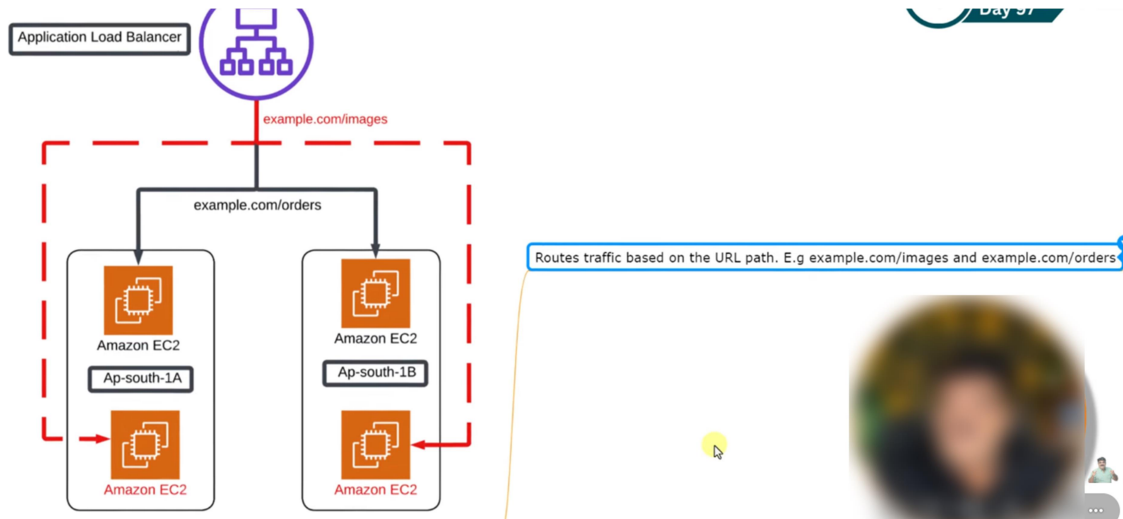
Route traffic to Info
Alias to Application and Classic Load Balancer ▼
Asia Pacific (Mumbai) ▼
Q dualstack.ALB-1630058658.ap-south-1.elb.amazonaws.com X
Alias hosted zone ID: ZP97RAFLXTNZK

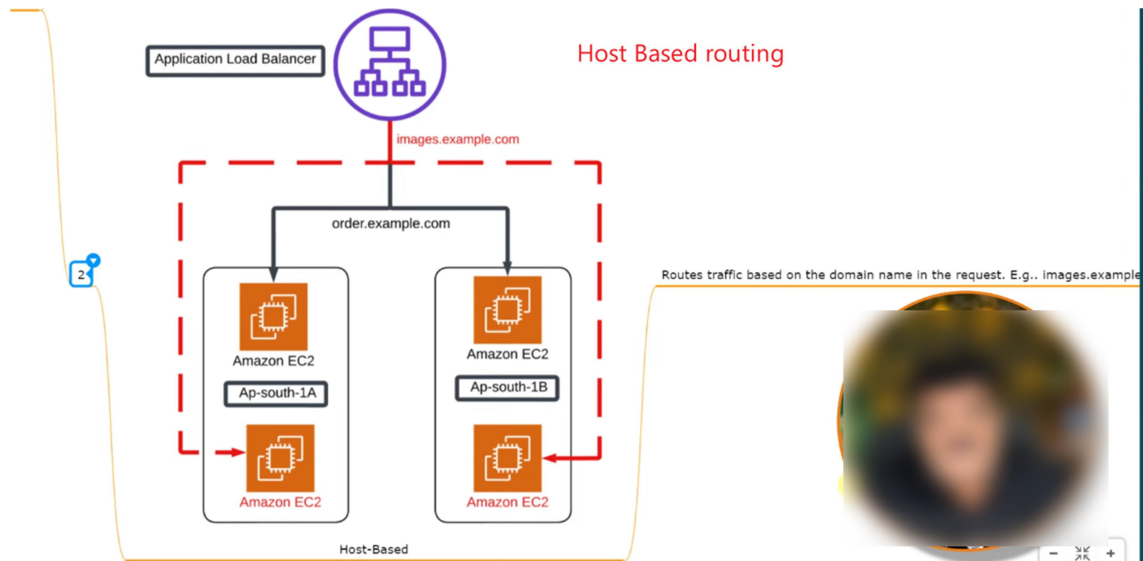
Routing policy Info
Simple routing ▼

Evaluate target health ☒ Yes

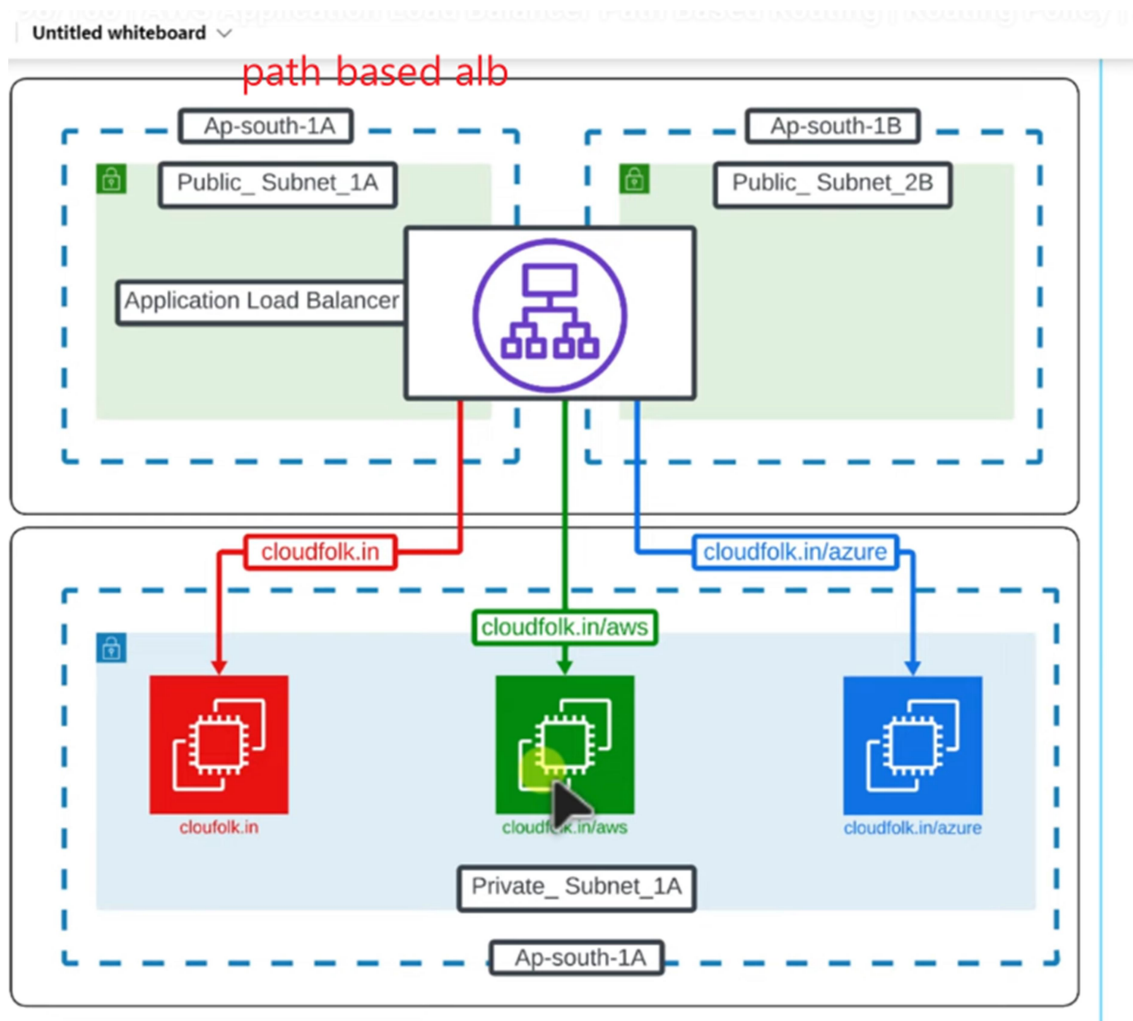
Path base and Host Based ALB:

Path Based routing





DNS Configuration	1	Path-Based	DNS configuration points to the ALB. No specific DNS setup for individual paths.
	2	Host-Based	Requires DNS records for each domain or subdomain to point to the ALB.
Use Cases	1	Path-Based	Useful for routing to different applications or services under the same domain.
	2	Host-Based	Ideal for routing to different applications or services based on different domains or subdomains.
SSL/TLS Certificates	1	Path Based	A single certificate can cover all paths under a domain.
	2	Host-Based	May require multiple certificates if different domains are used.
Flexibility	1	Path Based	Limited to routing within a single domain.
	2	Host-Based	More flexible, supporting multiple domains and subdomains.
Complexity	1	Path Based	Generally simpler to manage with fewer DNS records.
	2	Host-Based	Can be more complex due to managing multiple DNS entries and SSL certificates.



Create ALB>select myvpc>inbound traffic http anywhere.> create sg

Create web-sg>select myvpc>inbound http allow from alb-sg

Create 3 instance as per diagram name instance cloudfolks> ami> myvpc>private_subnet-1A>ipdisable>web-sg>

```

#!/bin/bash
yum update -y

# Install Apache
yum install -y httpd

# Start and enable Apache to run on boot
systemctl start httpd
systemctl enable httpd

# Create a simple HTML file in Root Directory
echo "This is the CloudFolks HUB Page served at Root" > /var/www/html/index.html
  
```

user datascrip paste

Create next aws instance same procedure as above instance but datascrip

```
#!/bin/bash
yum update -y

# Install Apache
yum install -y httpd

# Start and enable Apache to run on boot
systemctl start httpd
systemctl enable httpd

# Create a directory for /aws path
mkdir /var/www/html/aws

# Create a simple HTML file in the /var/www/html/aws/ Directory
echo "This is the AWS page served at /aws" > /var/www/html/aws/index.html
```

userdatascript for aws

Same for azure but datascrip copy above datascrip and some changes as below diag.

```
# Create a directory for /azure path
mkdir /var/www/html/azure

# Create a simple HTML file in the /var/www/html/azure/ Directory
echo "This is the Azure page served at /azure" > /var/www/html/azure/index.html
```

Create target group create target group> instance select > target group name :cloudfolk>ipv4 > vpc:myvpc>

Health checks
The associated load balancer periodically sends requests, per the settings below, to the registered targets to test their status.

Health check protocol
HTTP

Health check path
Use the default path of "/" to ping the root, or specify a custom path if preferred.
/

Up to 1024 characters allowed.

health check root means main page of application click next

Available instances (1/3)

Filter instances

	Instance ID	Name	State	Security groups
☐	i-06a115bff67d7a8e3	azure	Running	Web-SG
☐	i-07a4a43aad50a25b5	azure	Running	Web-SG
☒	i-02425c0e64c73f88b	cloudfolk	Running	Web-SG

1 selected

Ports for the selected instances
Ports for routing traffic to the selected instances.

80

1-65535 (separate multiple ports with commas)

Include as pending below

create target group

Next Target group for AWS > create target group> instance select > target group name :aws>ipv4 > vpc:myvpc>

Health check protocol

HTTP

/aws/

Health check path

Use the default path of "/" to ping the root, or specify a custom path if preferred.

/aws/

Up to 1024 characters allowed.

create next

Available instances (1/3)

Filter instances

	Instance ID	Name	State	Sec
☐	i-06a115bff67d7a8e3	azure	Running	Web
☒	i-07a4a43aad50a25b5	aws	Running	Web
☐	i-02425c0e64c73f88b	cloudfolk	Running	Web

1 selected

Ports for the selected instances
Ports for routing traffic to the selected instances.

80

1-65535 (separate multiple ports with commas)

Include as pending below

Third target group azure : create target group> instance select > target group name : azure>ipv4 > vpc:myvpc>

Health checks
The associated load balancer periodically sends requests, per the settings below, to the registered targets to test th

Health check protocol

HTTP

Health check path

Use the default path of "/" to ping the root, or specify a custom path if preferred.

/azure/

Up to 1024 characters allowed.

next

Available instances (1/3)

Filter instances

	Instance ID	Name	State	Security groups
☒	i-06a115bff67d7a8e3	azure	Running	Web-SG
☐	i-07a4a43aad50a25b5	aws	Running	Web-SG
☐	i-02425c0e64c73f88b	cloudfolk	Running	Web-SG

1 selected

Ports for the selected instances
Ports for routing traffic to the selected instances.

80

1-65535 (separate multiple ports with commas)

Include as pending below

create

Now create ALB

Go to ALB>create ALB> name:alb>ipv4>vpc:myvpc>

☒ **ap-south-1a (aps1-az1)**

Subnet

subnet-0b9acda39bc95d88b

Public_Subnet_1A

IPv4 address

Assigned by AWS

☒ **ap-south-1b (aps1-az3)**

Subnet

subnet-04a17b8f6a72da5e0

Public_Subnet_2B

IPv4 address

Assigned by AWS

Security group:alb-sg

▼ Listener HTTP:80 Remove

Protocol Port Default action Info

HTTP : 80

1-65535

Forward to cloudfolk HTTP

Target type: Instance, IPv4

Create target group

Listener tags - optional

Consider adding tags to your listener. Tags enable you to categorize your AWS resources so you can more easily manage them.

Add listener tag

You can add up to 50 more tags.

select cloudfolk t.g

and create load balancer > copy dns url and paste on browser check for root application

Next setup for aws and azure target go to load balancer and click created load balancer alb

Listeners and rules (1/1) [Info](#)

A listener checks for connection requests on its configured protocol and port. Traffic received by the listener is routed according to the default action. You can add additional rules.

☒	Protocol:Port	Default action	Rules	ARN	Security policy
☒	HTTP:80	Forward to target group cloudfolk: 1 (100%) - Group-level stickiness: Off	1 rule	ARN	Not applicable

Click to add rule

Name and tags [Info](#)

Tags can help you manage, identify, organize, search for and filter resources.

Name [Add additional tags](#)

[Cancel](#) [Next](#)

Conditions (0) [Rule limits](#)

No conditions
No conditions to display.

[Add condition](#)

[Cancel](#) [Previous](#) [Next](#)

Add condition

Rule limits

×

Rule condition types

Route traffic based on the condition type of each request. Each rule can include one of each of the following conditions: host-header, path, http-request-method and source-ip. Each rule can include one or more of each of the following conditions: http-header and query-string.

Path

Path

Define the path. For example: /item/*. Case sensitive.

is

/aws*

Maximum 128 characters. Allowed characters are a-z, A-Z, 0-9; the following special characters: _-\$/~'"@:~;~& (using &); and wildcards (* and ?).

Add new value

You can add up to 4 condition values for this rule.

Cancel

Confirm

next

Action types

Routing actions

☒ Forward to target groups

☐ Redirect to URL

☐ Return fixed response

Forward to target group

Info

Choose a target group and specify routing weight or [Create target group](#).

Target group

aws

Target type: Instance, IPv4

HTTP

↺

1

0-999

100%

Add target group

You can add up to 4 more target groups.

Rule: K1

Priority

Rule priority controls the evaluation order of a rule within the listener's set of rules.

100

1 - 50000

create

Now for azure : go to LB>select alb>listener and rule>select listener >manage rules>add rule>name:R2>add condition>select path base>

Add condition Rule limits

Rule condition types

Route traffic based on the condition type of each request. Each rule can include one of each of the following conditions: host-header, path, http-request-method and source-ip. Each rule can include one or more of each of the following conditions: http-header and query-string.

Path

Path

Define the path. For example: /item/*. Case sensitive.

is /azure*

Maximum 128 characters. Allowed characters: a-z, A-Z, 0-9; the following special characters: _-./~!@:~+; & (using &); and wildcards (* and ?).

Add new value

You can add up to 4 more condition values for this rule.

Cancel Confirm

Click confir and next

Target group

azure

Target type: Instance, IPv4

HTTP

Weight

Percentage

1

100%

0-999

Add target group

You can add up to 4 more target groups.

priority: 200

Next and create then go tar.group check for refresh target

now go to dns url on browser xyz/aws or xyz/azure